# Prayas JEE AIR (2025)

## WEEKLY TEST 20

**DURATION : 180 Minutes** | **DATE : 15/12/2024** | **M. MARKS : 300**

# ANSWER KEY

**PHYSICS**

1. (3)
2. (3)
3. (3)
4. (2)
5. (3)
6. (4)
7. (3)
8. (3)
9. (4)
10. (2)
11. (4)
12. (3)
13. (3)
14. (1)
15. (3)
16. (4)
17. (1)
18. (4)
19. (1)
20. (4)
21. (54)
22. (48)
23. (60)
24. (30)
25. (7)

**CHEMISTRY**

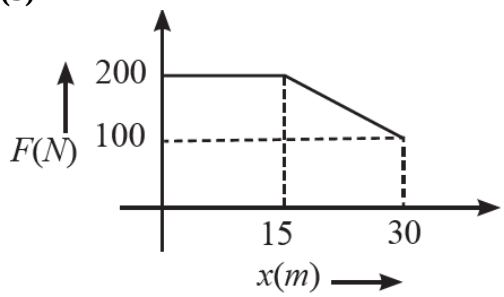
26. (3)
27. (4)
28. (4)
29. (1)
30. (1)
31. (3)
32. (3)
33. (2)
34. (1)
35. (4)
36. (1)
37. (2)
38. (1)
39. (2)
40. (3)
41. (4)
42. (4)
43. (3)
44. (3)
45. (4)
46. (12)
47. (1)
48. (20)
49. (2)
50. (2)

**MATHEMATICS**

51. (1)
52. (2)
53. (2)
54. (2)
55. (2)
56. (2)
57. (3)
58. (3)
59. (2)
60. (4)
61. (2)
62. (2)
63. (3)
64. (1)
65. (2)
66. (4)
67. (1)
68. (3)
69. (2)
70. (3)
71. (15)
72. (107)
73. (4)
74. (4)
75. (13)

## SECTION-I (PHYSICS)

**1.** **(3)**




Work = Area of $F$–$x$ curve

$$= 200\times 15 + \frac{1}{2}(200+100)\times 15 = 5250\text{J}$$

**2.** **(3)**

Power in external $R = \left(\frac{\varepsilon}{r+R}\right)^2 R$

This power is maximum, when $R = r$

$$r = \int_a^b \rho \frac{dx}{2\pi x l}$$

$$r = \frac{\rho}{2\pi l}\ln\frac{b}{a}$$

**3.** **(3)**

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = x$$

$$c = \frac{E}{B} = y$$

$$\tau = RC = t \Rightarrow \frac{I}{RC} = LT^{-1}$$

$\Rightarrow [x] = [y] = [z]$

**4.** **(2)**

$$q = \int I\,dt$$

$$q = \int_0^{L/R} \frac{E}{R}\left[1 - e^{\frac{-Rt}{L}}\right]dt \Rightarrow q = \frac{EL}{R^2}\frac{1}{e} \Rightarrow q = \frac{EL}{2.7R^2}$$

**5.** **(3)**

Moment of Inertia of bigger disc about axis $AB$,

$$I_1 = \frac{1}{2}MR^2 = \frac{1}{2}\sigma\pi R^2.R^2 \quad [\because M = \sigma\pi R^2]$$

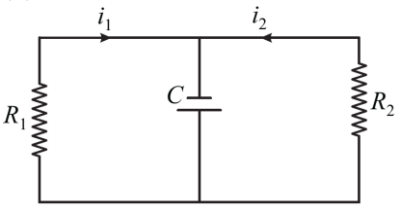
Moment of Inertia of bigger disc about axis $CD$,

$$I_2 = \frac{1}{4}mr^2 = \frac{1}{4}\sigma\pi r^2.r^2 \qquad [\because m = \sigma\pi r^2]$$

$$\therefore \frac{I_1}{I_2} = \frac{\left(\frac{1}{2}\sigma\pi R^2.R^2\right)}{\left(\frac{1}{4}\sigma\pi r^2.r^2\right)}$$

$$= \frac{2R^4}{r^4}$$

**6.** **(4)**




**7.** **(3)**

$$F = -\frac{dU}{dr} = -\left[\frac{6A}{r^7} - \frac{12B}{r^{13}}\right]$$

$F = 0$

$$\Rightarrow r = \left(\frac{2B}{A}\right)^{1/6}$$

$$U\left(at\, r = \left(\frac{2B}{A}\right)^{1/6}\right) = -\frac{A^2}{4B}$$

**8.** **(3)**

As length is same

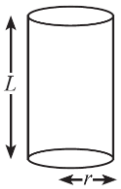
So

$$\frac{F}{\pi r^2} = Y\alpha T$$

linear coefficient of expansion $\alpha = \frac{F}{YT\pi r^2}$

$\therefore$ The coefficient of volume expansion $\lambda = 3\alpha$

$$\therefore \gamma = 3\frac{F}{YT\pi r^2}$$




**9.** **(4)**

A: For transition $A$, $n_1 = 1$, $n_2 = \infty \Rightarrow$ Series limit of Lyman series

B: For transition $B$, $n_1 = 2$, $n_2 = 5 \Rightarrow$ Third line of Balmer series

C: For transition $C$, $n_1 = 3$, $n_2 = 5 \Rightarrow$ Second line of Paschen series

**10.** **(2)**

$$W_{AB} = nRT\ln\frac{2V_1}{V_1} = nRT\ln 2$$

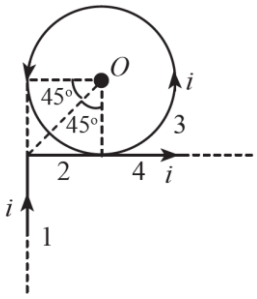
$$W_{BC} = P_2(V_1 - 2V_1) = -P_2V_1 = -\frac{1}{2}nRT$$

$[\because At\ B,\ 2P_2V_1 = nRT]$

As $CA$ is isochoric process, $W_{CD} = 0$

$$W_{ABCA} = W_{AB} = W_{BC} + W_{CA} = nRT\left(\ln(2) - \frac{1}{2}\right)$$

**11.** **(4)**




The net magnetic field

$$= \frac{\mu_0 i}{4\pi R}\left[\sin 90° - \sin 45°\right] \oplus + \frac{\mu_0 i}{2R} \odot +$$

$$= \frac{\mu_0 i}{4\pi R}(\sin 45° + \sin 90°) \odot$$

$$= \frac{-\mu_0 i}{4\pi R}\left[1 - \frac{1}{\sqrt{2}}\right] + \frac{\mu_0 i}{2R} + \frac{\mu_0 i}{4R}\left[\frac{1}{\sqrt{2}} + 1\right] \odot$$

$$= \frac{\mu_0 i}{4\pi R}\left[-1 + \frac{1}{\sqrt{2}} + 2\pi + \frac{1}{\sqrt{2}} + 1\right] \odot$$

$$= \frac{\mu_0 i}{4\pi R}\left[\sqrt{2} + 2\pi\right] \odot = \frac{\mu_0 i}{2\pi R}\left[\frac{1}{\sqrt{2}} + \pi\right] \odot$$

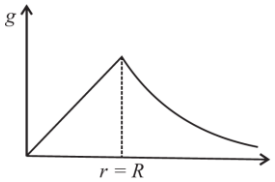
**12.** **(3)**

Acceleration due to gravity at height $h$ above the surface of earth.

$$g' = \frac{g}{\left(1 + \frac{h}{R}\right)^2}$$

Acceleration due to gravity at a depth $d$ below the surface of earth.

$$g' = g\left\{1 - \frac{d}{R}\right\}$$




**13.** **(3)**

$$E.4\pi r^2 = \frac{\int_0^a (kr).4\pi r^2 dr}{\varepsilon_0}$$

$$\Rightarrow E = \frac{kr^2}{4\varepsilon_0} \qquad \ldots. (i)$$

$$\text{Now, } 2Q = \int_0^R kr.4\pi r^2 dr \Rightarrow K = \frac{2Q}{\pi R^4} \qquad \ldots (ii)$$

From (i) and (ii)

$$E = \frac{Qr^2}{2\pi\varepsilon_0 R^4}$$

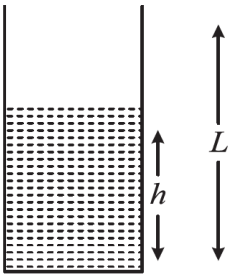
According to Question

$F_{net} = 0$

$$\Rightarrow \frac{Q_2}{4\pi\varepsilon_0 (2a)^2} = EQ$$

$a = R.8^{-1/4}$.

**14.** **(1)**




$$H - h_1 = n\frac{\lambda}{2} + \frac{\lambda}{4}$$

$$L - h_2 = (n-1)\frac{\lambda}{2} + \frac{\lambda}{4}$$

$$\Rightarrow \frac{\lambda}{2} = h_2 - h_1 = 24.5 - 17.0 = 7.5\,\text{cm}$$

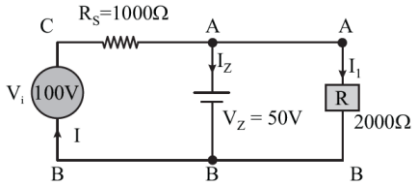
$\Rightarrow \lambda = 15$ cm

$u = f\lambda$

$f = 2200$ Hz

**15.** **(3)**

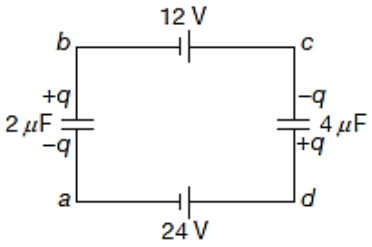
$$I_2 = \frac{V_i - V_z}{R_s} - \frac{V_z}{R} = \frac{100 - 50}{1000} - \frac{50}{2000} = 25\,\text{mA}$$




**16.** **(4)**

Apply Kirchhoff's Law

$$V_a = \frac{q}{2} + 12 + \frac{q}{4} - 24 = V_a$$




$$\frac{3}{4}q = 12$$

$\Rightarrow \quad q = 16\ \mu C$

$$V_a + \frac{16}{2} = V_b$$

$\Rightarrow \quad V_a - V_b = -8$ V

Similarly $V_d - V_c = +4$ V

Work done by cell 12 V is $-12 \times 16 = -192$ μJ

Work done by cell 24 V is $24 \times 16 = 384$ μJ

**17.** **(1)**

$$K.E. = \frac{hc}{\lambda} - \phi$$

As radius, $r = \frac{mv}{eB} = \frac{\sqrt{2(K.E)m}}{eB}$

$\Rightarrow (reB)^2 = 2m\,(K.E)$

$$\Rightarrow K.E = \frac{e^2B^2r^2}{2m} = \frac{(1.6^{-19} \times 2 \times 10^{-3} 2 \times 10^{-3})^2}{2 \times 9.1 \times 10^{-31}}$$

$= 2.25 \times 10^{-19}$J

$$= \frac{2.25 \times 10^{-19}}{1.6 \times 10^{-19}} \text{eV} = 1.40\,\text{eV}$$

Work function,

$$\phi = \frac{hc}{\lambda} - K.E.$$

$$= \frac{12400}{4500} - 1.40$$

$= 2.76 - 1.40$

$= 1.36$ eV

**18.** **(4)**

In the initial position, the length of the stretched spring is

$$L = \frac{h}{\cos 37°} = \frac{5h}{4} = 1.25h$$

So, extension in the spring is

$x = L - h = 0.25\,h$

*PE* stored in the spring is

$$U = \frac{1}{2}kx^2 = \frac{1}{2}k(0.25h)^2 = \frac{kh^2}{32}$$

When the spring becomes vertical, it acquires natural length and hence the PE gets converted into kinetic energy of the ring

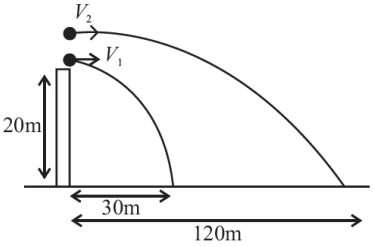
$$\Rightarrow \frac{1}{2}mv^2 = \frac{kh^2}{32}$$

$$\Rightarrow v = \frac{h}{4}\sqrt{\frac{k}{m}}$$

**19.** **(1)**

Refractive index is changing, the light cannot travel in a straight line in the liquid.

**20.** **(4)**




$$v_1 = \frac{30}{\sqrt{\frac{2h}{g}}},\ v_2 = \frac{120}{\sqrt{\frac{2h}{g}}}$$

$$(0.01)u = (0.2)\frac{30\sqrt{g}}{\sqrt{2h}} + (0.01)\frac{120\sqrt{g}}{\sqrt{2h}}$$

$u = 300 + 60 = 360\ \text{ms}^{-1}$

**21.** **(54)**

$a = -\mu_k g = -3$

$V = 18 - 3 \times 2$

$V = 12$ m/s

$$KE = \frac{1}{2}mv^2 + \frac{1}{2}\frac{mr^2}{2}\frac{v^2}{r^2} = \frac{3}{4}mv^2$$

$KE = 3 \times 15 = 54$J

**22.** **(48)**

It is given,

Wavelength of 1st wave $\lambda_1 = 800$ nm

Wavelength of 2nd wave $\lambda_2 = 600$ nm

Distance of screen from the slit, $D = 7$m

Distance between the two slits, $d = 0.35$ mm

$$n_1\frac{\lambda_1 D}{d} = n_2\frac{\lambda_2 D}{d}$$

$$800n_1 = 600n_2 \Rightarrow \frac{n_1}{n_2} = \frac{3}{4}$$

Minimum $n_1 = 3$ & $n_2 = 4$

$$\text{Minimum separation} = \frac{n_1\lambda_1 D}{d}$$

$$= \frac{3 \times 800nm \times 7m}{0.35mm} = 48$$

**23.** **(60)**

$$2\sin\theta c = \sqrt{3}$$

$$\text{So, } \sin\theta c = \frac{\sqrt{3}}{2}$$

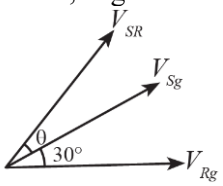
$$\theta c = 60°$$

**24.** **(30)**

Both velocity vector are of same magnitude. Therefore resultant would pass exactly midway through them.

So, $\theta = 30°$

Hence, angle $\theta$ with the line *AB* is 30°




**25.** **(7)**

Velocity of second ball

$$\left(\frac{m/2 - m}{m/2 + m}\right)(0) + \left(\frac{2 \times m}{m + m/2}\right)v$$

$$= \frac{4}{3}v$$

Velocity of third ball will become,

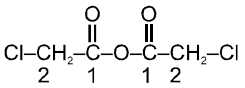
$$\frac{4}{3}\text{times } \frac{4}{3}v \text{ or } \left(\frac{4}{3}\right)^2 v$$

$$\therefore v_n = \left(\frac{4}{3}\right)^{n-1} v = \sqrt{5gr}$$

$$\therefore v = \left(\frac{3}{4}\right)^{n-1}\sqrt{5gr}$$

## SECTION-II (CHEMISTRY)

**26. (3)**



Chloroacetic anhydride or chloroethanoic anhydride

**27. (4)**

N and O have ability to form pπ-pπ multiple bonds with itself on account of smaller size of atoms. N – N and O – O bond energies are less on account of repulsion between non-bonded pairs of electrons due to smaller size of atoms. S – S bond energy (265 kJ $mol^{-1}$) is next to C – C

**28. (4)**

The d-subshells are not filled with electrons monotonically with increase in atomic number. There are some exceptions like Cr, Cu etc.

**29. (1)**

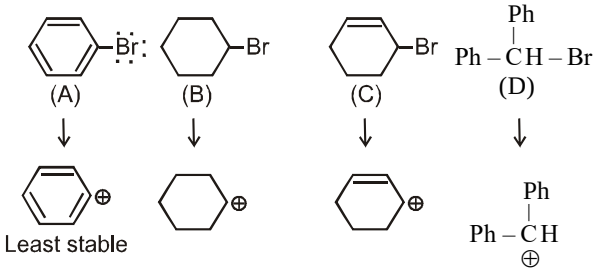
$$\text{Molality} = \frac{x_A \times 1000}{(1-x_A)m_B} \qquad \left\{\begin{matrix} A \rightarrow \text{solute} \\ B \rightarrow \text{solvent} \end{matrix}\right\}$$

$$\text{First solution's molality} = \frac{0.25\times1000}{0.75\times18}\ \text{m}$$

$$\text{Second solution's molality} = \frac{0.5\times1000}{0.5\times18}\ \text{m}$$

$$\therefore \quad \text{Molality ratio} = \frac{1}{3}.$$

**30. (1)**

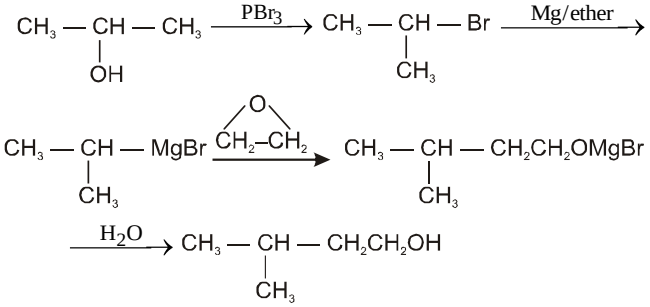



2° carbocation (stabilized by hyperconjugation and + I effect)

Allylic carbocation is stabilized by resonance and hyper – conjugation

Resonance stabilized

**31. (3)**



**32. (3)**

$$P_T = X_A \cdot P_A^* + X_B \cdot P_B^* = P_A + P_B$$

$$P_A = Y_A \cdot P_T \text{ and } P_B = Y_B \cdot P_T$$

**33. (2)**

Here $n_2 = 5$ & $n_1 = 2$

So longest wavelength means least energy difference transition i.e., $n_2 = 5$ to $n_1 = 4$

$$\frac{1}{\lambda} = R_H(1)^2\left(\frac{1}{(4)^2} - \frac{1}{(5)^2}\right)$$

So $\lambda = 4 \times 10^{-6}$ m

**34. (1)**

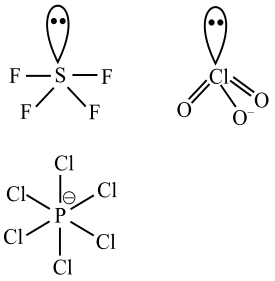
$Fe^{3+} \rightarrow 3d^5$ (Paramagnetic) and

$Fe^{2+} \rightarrow 3d^6$ Diamagnetic.

**35. (4)**

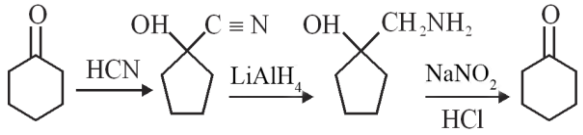
$PCl_6^- - sp^3d^2 - \text{Octahedral}$

$^-N = \overset{+}{N} = N^-$



**36. (1)**

Here the reaction occurs as follows to give cyclohexanone




**37. (2)**

Prussian blue colour is obtained due to the formation of $\left[Fe_4\left(Fe(CN)_6\right)_3\right]$

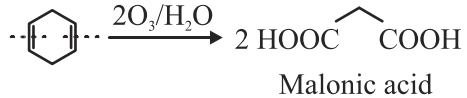
**38. (1)**

Order of bond length

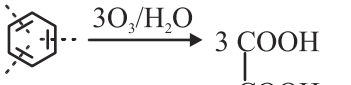
$BaO_2 > KO_2 > O_2 > O_2[PtF_6]$

BO : (1) (1.5) (2) (2.5)

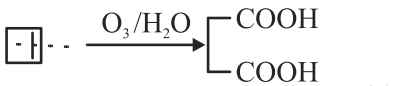
**39. (2)**

(i) 


Malonic acid

(ii) 

Oxalic acid

(iii) 

succinic acid

**40. (3)**

Aldehydes give 2° Alcohols on reaction with grignard Reagent.

**41. (4)**

Orthohydroxy benzoic acid is more acidic than orthonitro phenol.

**42. (4)**

For ppt $K_{ip} > K_{sp}$

$[IO_3^-] = 0.01$

$[Cu^{2+}] = 0.001$

$Cu(IO_3)_2 \rightleftharpoons Cu^{2+} + 2IO_3^-$

$K_{ip} = [Cu^{2+}]^1[IO_3^-]^2$

$K_{ip} = (0.001)\ (0.01)^2 = 10^{-7}$

$K_{ip} > K_{sp}$, ppt occur.

**43. (3)**

Reactions that occurs at anode

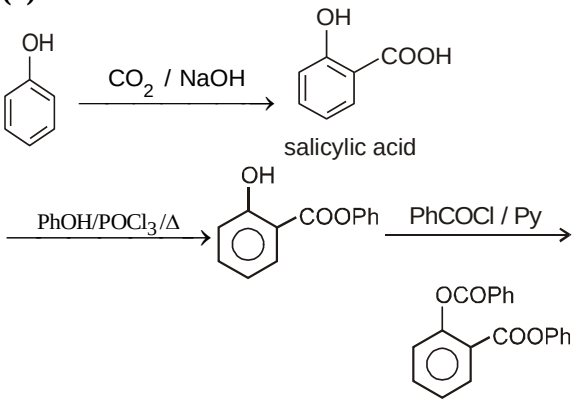
$Zn \rightarrow Zn^{2+} + 2e^-$

At cathode

$Br^2 \rightarrow 2Br^- + 2e^-$

$\therefore$ $[Zn^{2+}]$ and $[Br^-]$ increases and $[Br_2]$ and mass of Zn rod decreases with time.

**44. (3)**




**45. (4)**

$$\underset{(S_2)}{2K_2Cr_2O_7} \xrightarrow{\Delta} O_2 + 2K_2Cr_2O_4 + Cr_2O_3$$

$$\underset{(S_1)}{(NH_4)_2Cr_2O_7} \xrightarrow{\Delta} N_2 + Cr_2O_3 + H_2O$$

**46. (12)**

$$W = -nRT \ln\left(\frac{P_1}{P_2}\right)$$

$= 5 \times 2 \times 300 \times \ln(50)$

$= 5 \times 2 \times 300 \times 2.303 \times (1 + 0.7)$

$= 11.745$ kcal $\approx$ 12 kcal

**47. (1)**

$CoCl_3 . 4NH_3 \Rightarrow [Co(NH_3)_4Cl_2]Cl$ (G.I.)

$CoCl_3 . 5NH_3 \Rightarrow [Co(NH_3)_5Cl]Cl_2$ (No G.I.)

$CoCl_3 . 6NH_3 \Rightarrow [Co(NH_3)_6]Cl_3$ (No G.I.)

$CoCl.(NO_3)_2.5NH_3 \Rightarrow [Co(NH_3)_5Cl]\ (NO_3)_2$ (No G.I.)

Only $[Co(NH_3)_4Cl_2]$ can show geometrical isomerism. Hence it can exist in cis-trans form.

**48. (20)**

$Br_2(l) \rightleftharpoons Br_2(g)$

At equilibrium $\underset{1-x}{Br_2(l)} + Cl_2(g) \rightleftharpoons \underset{2x}{2BrCl(g)}$

$$K_P = \frac{(P_{BrCl})^2}{P_{Cl_2}} = \frac{(P_{BrCl})^2}{0.01} = 1$$

$\Rightarrow$ $P_{BrCl} = 0.1$ atm

$$\frac{n_{BrCl}}{n_{Cl_2}} = \frac{0.1}{0.01} = 10 = \frac{2x}{1-x}$$

$$\Rightarrow \quad x = \frac{10}{12} = \frac{5}{6}$$

$\therefore$ Moles of $Br_2(l)$ reacted with $Cl_2(g) = \frac{5}{6}$ moles.

$\because$ Vapour pressure of $Br_2(l)$ = Partial pressure of BrCl(g)

$\therefore$ Moles of $Br_2(l)$ converted into vapour = $\frac{10}{6}$

$\therefore$ Minimum moles of $Br_2(l)$ required

$$= \frac{10}{6} + \frac{5}{6}$$

$$= \frac{15}{6}$$

$\therefore$ Mass $= \frac{15}{6} \times 160$

$= 400$ g

**49. (2)**

Molarity of $H_2O_2 = \frac{20}{11.2} = 1.786$

For titration, $[M_1V_1]\ V.F_1 = [M_2V_2]\ V.F_2$

$M_1 \times 10 \times 2 = 0.025 \times 25 \times 5$

$M_1 = 0.15625$

Now molarity of diluted solution = 0.15625

So molarity of original solution = 1.563

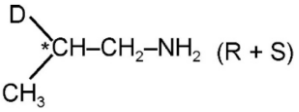
So, $K = \frac{1}{t} \ln\left(\frac{C_0}{C_t}\right)$

$$K = \frac{2.303}{6} \log\left(\frac{1.786}{1.563}\right)$$

**50. (2)**

There will be no intermolecular reaction between amides

Product obtained are



## SECTION-III (MATHEMATICS)

**51. (1)**

$$\frac{(e^x-\sin x)(\cos x-3)(\ln x+2)}{\left(x^4-x+\frac{1}{2}\right)(2x-7)^{30}}\le 0$$

$$x^4-x+\frac{1}{2}=\left(x^4-x^2+\frac{1}{4}\right)+\left(x^2-x+\frac{1}{4}\right)$$

= positive

Using wavy curve $x\in\left[\frac{1}{e^2},\frac{7}{2}\right)\cup\left(\frac{7}{2},\infty\right)$.

**52. (2)**

We have $J=\int_{-5}^{-4}(3-x^2)\tan(3-x^2)dx.$

Put $(x + 5) = t$, we get

$$J=\int_0^1(3-(t-5)^2)\tan(3-(t-5)^2)dt$$

$$=\int_0^1(-22+10t-t^2)\tan(-22+10t-t^2)dt$$

Now, $K=\int_{-2}^{-1}(6-6x+x^2)\tan(6x-x^2-6)dx.$

Put $(x + 2) = z$, we get

$$K=\int_0^1(6-6(z-2)+(z-2)^2)\tan(6(z-2)-(z-2)^2-6)dz$$

$$\int_0^1(22-10z+z^2)\tan(-22+10z-z^2)dz$$

Hence (J + K) = 0

**53. (2)**

$\sin^{-1}\left(\frac{5}{x}\right)$ and $\sin^{-1}\left(\frac{12}{x}\right)$ are both defined when $x \ge 12$. Therefore

$$\cos^{-1}\left(\sqrt{1-\frac{25}{x^2}}\right)+\sin^{-1}\left(\frac{12}{x}\right)=\frac{\pi}{2}$$

$$\left\{\because \sin^{-1}\frac{2}{x}+\cos^{-1}\frac{2}{x}=\frac{\pi}{2}\right\}$$

$$\sin^{-1}\left(\frac{12}{x}\right)=\sin^{-1}\sqrt{1-\frac{25}{x^2}}$$

On comparing, we get $\frac{12}{x}=\sqrt{1-\frac{25}{x^2}}$

Therefore $144 = x^2 - 25$ or $x = 13$

**54. (2)**

Centre of rectangular hyperbola is (1, –2)

So, equation of asymptotes are $x = 1$, $y = -2$

Hence, radius of circle = $\frac{\sqrt{5}}{2}$

**55. (2)**

$$T_{r+1}={}^{16}C_r\cdot\left(\frac{x}{\cos\theta}\right)^{16-r}\cdot\left(\frac{1}{x\sin\theta}\right)^r$$

$$={}^{16}C_r\cdot\frac{x^{16-2r}}{(\cos\theta)^{16-r}\cdot(\sin\theta)^r}$$

$\because 16 - 2r = 0 \Rightarrow r = 8$

$$T_9=\frac{{}^{16}C_8\cdot 2^8}{\sin^8 2\theta}$$

$\because \sin 2\theta$ is increasing in $\left[\frac{\pi}{16},\frac{\pi}{4}\right]$

Hence, $a=\frac{{}^{16}C_8\cdot 2^8}{\sin^8\left(\frac{\pi}{2}\right)}$ and $b=\frac{{}^{16}C_8\cdot 2^8}{\sin^8\left(\frac{\pi}{4}\right)}$

$a + b = {}^{16}C_8\, 2^8 \cdot 17$

**56. (2)**

We have $|(A^{-1})\ \text{adj}(B^{-1})\ \text{adj}(2A^{-1})| =$

$$\frac{1}{|A|}\cdot\frac{1}{|B|^2}\cdot\frac{64}{|A|^2}=\frac{64}{-2\times1\times4}=-8$$

(By using $|(A^{-1})| = \frac{1}{|A|}$ and $|\text{adj } B| = |B|^{n-1}$ where $n$ is order of square matrix)

**57. (3)**

$$\int\frac{9\cos 2x+12\cos x-6\sin x}{(2+3\sin x)(4+3\cos x)}\,dx=$$

$$\int\frac{3\cos x(4+3\cos x)+(2+3\sin x)(-3\sin x)}{(2+3\sin x)(4+3\cos x)}dx$$

$$=\int\frac{3\cos x}{2+3\sin x}dx+\int\frac{-3\sin x}{4+3\cos x}dx$$

$= \ln(2+3 \sin x) + \ln (4 + 3 \cos x) + c$

$= \ln ((2+3 \sin x) (4+3 \cos x) + c$

**58. (3)**

$$2^{\frac{1}{4}}\cdot4^{\frac{1}{16}}\cdot8^{\frac{1}{48}}\cdot.....$$

$$=2^{\frac{1}{4}}\cdot2^{\frac{1}{8}}\cdot2^{\frac{1}{16}}\cdot.....$$

$$=2^{\left(\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+.....\right)}$$

$$=2^{\frac{1/4}{1-1/2}}=2^{\frac{1}{2}}$$

$x^2 - x = 2$

$x = 2, -1$

**59. (2)**

$$\tan(\theta_1+\theta_2+\theta_3)=\frac{\sum\tan\theta_1-\prod\tan\theta_1}{1-\sum\tan\theta_1\tan\theta_2}$$

$$=\frac{p+1-q}{1-(q-p)}=1$$

**60. (4)**

The symmetric difference of sets $A$ and $B$, denoted by $A \Delta B$, is given by:

$A \Delta B = (A \cup B) - (A \cap B)$

First, find $A \cup B$ and $A \cap B$:

$A \cup B = \{1, 2, 3, 4, 5\}$

$A \cap B = \{3\}$

Now, find the symmetric difference:

$A \Delta B = (A \cup B) - (A \cap B)$

$A \Delta B = \{1, 2, 3, 4, 5\} - \{3\}$

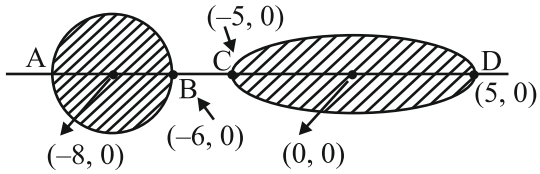
$A \Delta B = \{1, 2, 4, 5\}$

**61. (2)**

AD is maximum distance

BC is minimum distance

Range is [1, 15]




**62. (2)**

$$f'(x)=\frac{x^6}{6}-\frac{12}{15}x^5+\frac{15x^4}{20}+\frac{4x^3}{3}-2x^2+99$$

$f''(x) = x^5 - 4x^4 + 3x^3 + 4x^2 - 4x = 0$

$x(x^4 - 4x^3 + 3x^2 + 4x - 4) = 0$

$x(x - 1)(x - 2)^2 (x + 1) = 0$

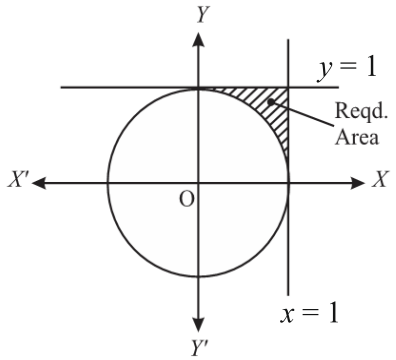
Pt of inflection at $x = -1, 0$ & $1$

So $-1 + 0 + 1 = 0$

**63. (3)**

By definition the lines max, $(x, y)$ = means.

$x = 1$ and $y \le 1$ or $y = 1$ *and* $x \le 1$




Required area $= \int_0^1 \left[1-\sqrt{1-x^2}\right]dx$

$$=\left[x-\frac{x}{2}\sqrt{1-x^2}-\frac{1}{2}\sin^{-1}x\right]_0^1=1-\frac{\pi}{4}$$

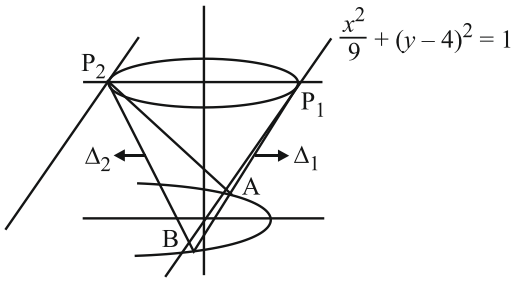
Or

Required area = $1^2-\frac{1}{4}\pi(1)^2 = 1-\frac{\pi}{4}$

**64. (1)**

$\Rightarrow \Delta_1 = 2\sqrt{2}(4-\sqrt{10})$

and $\Delta_2 = 2\sqrt{2}(4+\sqrt{10})$




**65. (2)**

$f(n) = (1)^\infty$

apply the formula

$$f(n)=\lim_{x\to0}\ e^{\frac{1}{x}\left(\left(1+\sin\frac{x}{2}\right)\left(1+\sin\frac{x}{2^2}\right)....\left(1+\sin\frac{x}{2^n}\right)-1\right)}$$

$$=\lim_{x\to0}\ e^{\frac{\left(1+\left(\sin\frac{x}{2}+\sin\frac{x}{2^2}+...+\sin\frac{x}{2^n}\right)+\left(\sin\frac{x}{2}.\sin\frac{x}{2^2}+...\right)-1\right)}{x}}$$

$$=\lim_{x\to0}\ e^{\left(\frac{\sin\frac{x}{2}}{x}+\frac{\sin\left(\frac{x}{2^2}\right)}{x}+....+\frac{\sin\frac{x}{2^n}}{x}\right)}=$$

$$e^{\left(\frac{1}{2}+\frac{1}{2^2}+....\frac{1}{2^n}\right)}$$

sum of infinite G.P. = $e^{\frac{\frac{1}{2}}{\left(1-\frac{1}{2}\right)}} = e$.

**66. (4)**

$\vec{c} = \alpha(\hat{i}+\hat{j}) + \beta(\hat{j}+k)$

$= \alpha\hat{i} + (\alpha+\beta)\hat{j} + \beta k$

$\because \vec{c}, \hat{i}-2j+k, 3\hat{i}+2j-k$ are coplanar

$$\therefore \begin{vmatrix} \alpha & \alpha+\beta & \beta \\ 1 & -2 & 1 \\ 3 & 2 & -1 \end{vmatrix} = 0$$

$\alpha(2 - 2) - (\alpha + \beta)(-1 - 3) + \beta(2 + 6) = 0$

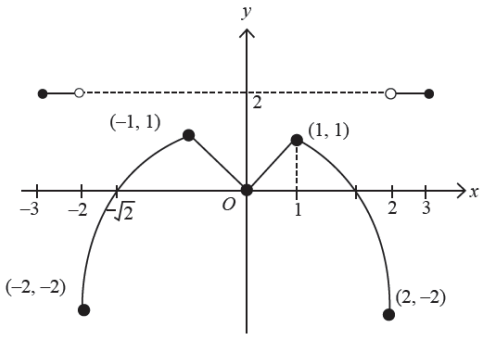
$4\alpha + 4\beta + 8\beta = 0$

$$\frac{\alpha}{\beta} = -3$$

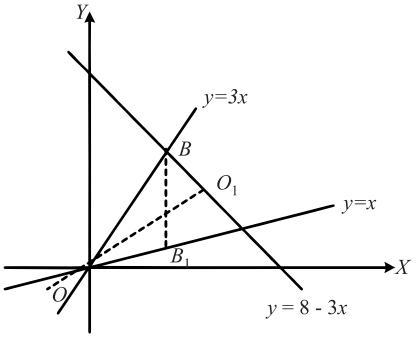
**67. (1)**

$$f(x)=\begin{cases}\min\{|x|,2-x^2\}, & -2\le x\le 2\\ [|x|], & 2<|x|\le 3\end{cases}$$

Number of points $f$ is not differentiable = 5.
Number of points $f$ is not continuous = 2.




**68. (3)**




Equation of bisector $OO_1$ is $\frac{y-3x}{\sqrt{10}}=\pm\frac{y-x}{\sqrt{2}}$ Here we will get two equation of the bisectors, but we have to consider the bisector which is bisecting the internal angle of the triangle. For the internal bisector the slope of the bisector should be positive.

$$\Rightarrow\left(\frac{1}{\sqrt{5}}+1\right)y-x\left(1+\frac{3}{\sqrt{5}}\right)=0 \quad \ldots.(1)$$

equation of $BB_1$ is

$$\frac{y+3x-8}{\sqrt{10}}=\frac{y-3x}{\sqrt{10}}\Rightarrow x=\frac{4}{3} \quad \ldots..(2)$$

$$\Rightarrow \text{incentre}=\left(\frac{4}{3},\left(\frac{3+\sqrt{5}}{1+\sqrt{5}}\right)\frac{4}{3}\right)=\left(\frac{4}{3},\frac{2+2\sqrt{5}}{3}\right)$$

**69. (2)**

Equation of line $L$ is $\vec{r}=(\hat{j}-\hat{k})+\lambda\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 1 & 4\\ 3 & 2 & -2\end{vmatrix}$

$\vec{r}=(\hat{j}-\hat{k})+\lambda(-10\hat{i}+16\hat{j}+\hat{k})$

Point $Q$ is $(-10\lambda, 16\lambda+1, \lambda-1)$

$|\overrightarrow{PQ}|^2=357\lambda^2=357$

$\lambda^2=1 \Rightarrow \lambda=\pm1$

$Q\equiv(-10, 17, 0)$ or $(10, -15, -2)$

then $a+2b+3c=24$ or $-26$

**70. (3)**

$$\begin{bmatrix}a_{11} & a_{12}\\ a_{21} & a_{22}\end{bmatrix}\begin{bmatrix}1 & 1\\ 2 & 1\end{bmatrix}=\begin{bmatrix}1 & 1\\ 2 & 1\end{bmatrix}\begin{bmatrix}a_{11} & a_{12}\\ a_{21} & a_{22}\end{bmatrix}$$

$\Rightarrow a_{11}+2a_{12}=a_{11}+a_{21}$ …(i)

$a_{11}+a_{12}=a_{12}+a_{22}$ …(ii)

$a_{21}+2a_{22}=2a_{11}+a_{21}$

$a_{21}+a_{22}=2a_{12}+a_{22}$

$\therefore$ From (i) & (ii), $a_{11}=a_{22}$

$a_{21}=2a_{12}$

$\because$ Given matrix is singular

$\Rightarrow a_{11}a_{22}=a_{12}a_{21}$

$\Rightarrow (a_{11})^2=2(a_{12})^2$

$\Rightarrow a_{11}=\sqrt{2}a_{12}$

**71. (15)**

$$xdy=\left(\sqrt{16x^2+y^2}+y\right)dx$$

Given, $\frac{dy}{dx}=\frac{y}{x}+\frac{\sqrt{y^2+16x^2}}{x}$

Put $y = V \cdot x$, in the above eq$^n$ and

$\frac{dy}{dx}=V+x\cdot\frac{dV}{dx}$, we get

$$\Rightarrow \quad V+x\frac{dV}{dx}=V+\sqrt{V^2+16}$$

$$\Rightarrow \quad x\frac{dV}{dx}=\sqrt{V^2+16}$$

$$\Rightarrow \quad \int\frac{dV}{\sqrt{V^2+16}}=\int\frac{dx}{x}+\ell nC$$

$$\Rightarrow \quad \ell n\left|V+\sqrt{V^2+16}\right|=\ell nCx$$

$$\Rightarrow \quad \frac{y}{x}+\frac{\sqrt{y^2+16x^2}}{x}=Cx$$

Given $y(1) = 3 \Rightarrow C = 8$

Now at $x = 2$

$$\Rightarrow \quad \frac{y}{2}+\frac{\sqrt{y^2+16\cdot4}}{2}=8\cdot2$$

$$\Rightarrow \quad y+\sqrt{y^2+64}=32$$

$\Rightarrow y^2+64=(32-y)^2$

$= y^2+64=y^2-64y+32^2$

$\Rightarrow y = 15$

**72. (107)**

Player $A$ can win if $A$ throws (1, 6) or (6, 1) and $B$ throws ((1, 1,), (2, 2), (3, 3), (4, 4), (5, 5) or (6, 6)). Thus the number of ways is 12.

Similarly the number of ways in which $B$ can win is 12.

Total number of ways in which either $A$ wins or $B$ wins = 24.

Thus the number of ways in which none of the two wins = $6^4 - 24$.

$\therefore$ The required probability = $\frac{6^4 - 24}{6^4} = \frac{53}{54}$.

**73. (4)**

$K = {}^9C_3 \times {}^3C_2 \times \frac{5!}{2!\ 2!\ 1!}$

**74. (4)**

Converting tanα and cotα in sinα and cosα:

$y = \sqrt{2\cot\alpha + \text{cosec}^2\alpha} = \sqrt{2\cot\alpha + 1 + \cot^2\alpha}$

$= |\cot\alpha + 1|$

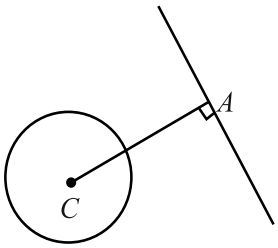
As $\alpha \in \left(\frac{3\pi}{4}, \pi\right)$

$y = -1 - \cot\alpha$

$\frac{dy}{d\alpha} = \text{cosec}^2\alpha$

At $\alpha = \frac{5\pi}{6}$, $\frac{dy}{d\alpha} = 4$

**75. (13)**




$C(-2,-3); r = \sqrt{4+9+5} = 3\sqrt{2}$

Length of perpendicular from centre to given line

$P = \left|\frac{-2-3+13}{\sqrt{2}}\right| = 4\sqrt{2}$

Clearly $p > r$, which implies that the line is away from given circle. (no intersection)

For nearest point, $A$ = foot of perpendicular from the centre $C$ (–2, –3) to the line.

$\frac{x+2}{1} = \frac{y+3}{1} = -\frac{(8)}{2} = -4$, we get $A(-6, -7)$



**PW Web/App** - https://smart.link/7wwosivoicgd4

**Library-** https://smart.link/sdfez8ejd80if